Qwen3.8-27B Agentic Search Eval on Private Data

A head-to-head replacement decision against Qwen3.6-35B-A3B, on one NVIDIA L4, over a corpus that never leaves the premises.

Han Xiao
VP of AI, Elastic

@hxiao · in/hxiao87

THIS DECK

The replacement decision

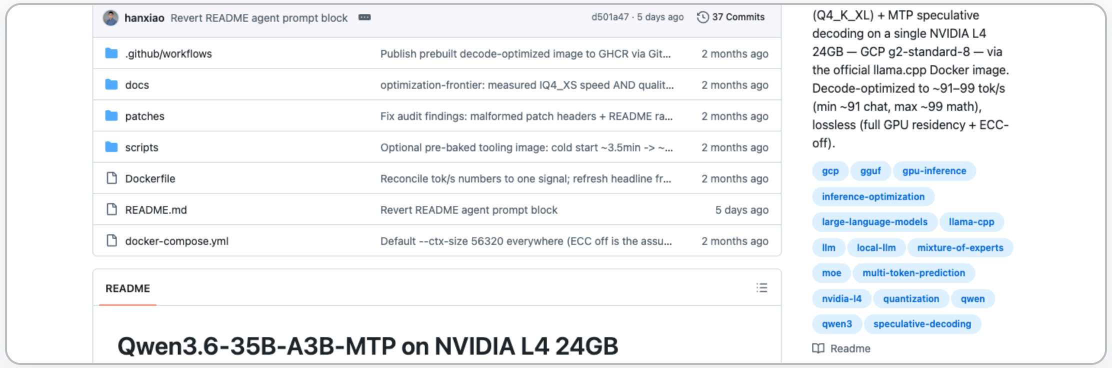
Qwen3.8 was released last week and has been widely promoted since. For agentic search on my own hardware I serve **Qwen3.6-35B-A3B** by default, and I have spent months tuning how it is served. The public reception of a new model does not tell me whether to replace it.

> **In agentic search over my own documents, how much better is Qwen3.8-27B than Qwen3.6-35B-A3B?**
>
> `the question this evaluation answers`

I evaluate two specific backends that I have already deployed and measured on the same card. A public leaderboard cannot answer the question, because the documents I search are private and the evaluation has to be built from them.

The two serving backends

Qwen3.6-35B-A3B-MTP



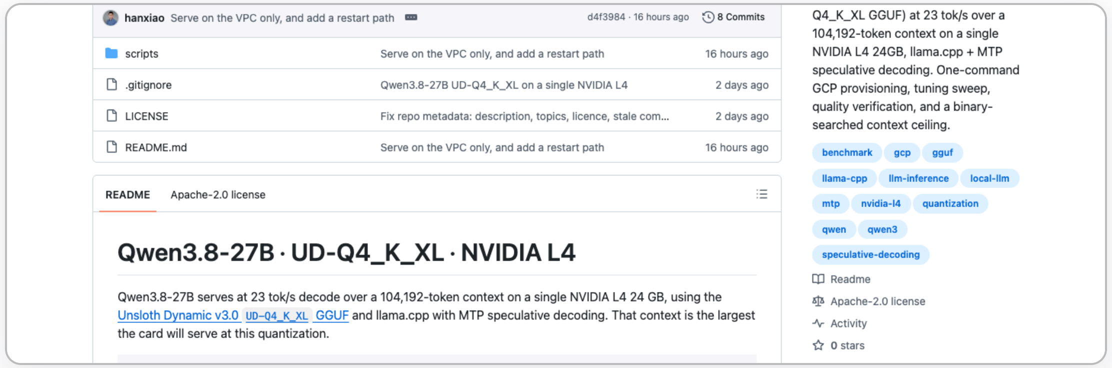
sparse MoE · 35B total, ~3B active · 8 of 256 experts per token · UD-Q4_K_XL ~21.3 GiB resident · MTP head baked in

91–99
tok/s decode, chat minimum to math maximum

56,320
token context

Per workload: prose 92.5, code 93.0, json 92.0, chat 90.6, math 98.8, translation 91.7, summarization 93.1 tok/s. That is 45% over the 63 tok/s out-of-the-box configuration, with full GPU residency of the experts and ECC off. Raw ceiling ~73 tok/s, which MTP multiplies 1.27 to 1.37 times.

Qwen3.8-27B-UD-Q4_K_XL



hybrid attention · 64 layers, full_attention_interval 4, so 16 hold a KV cache · native MTP head · positions to 262,144

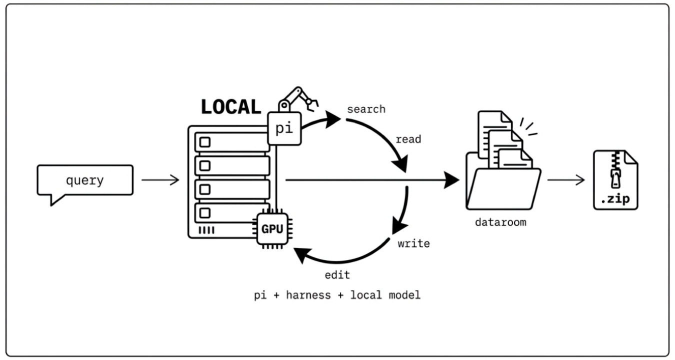
23
tok/s decode, binary-searched ceiling

104,192
token context, the largest the card serves at this quantization

At 98% of the window a 101,815-token prompt returned a correct answer at 24.88 tok/s decode and 278 tok/s prefill, with VRAM peaking at 24,082 MiB. Short-prompt prefill runs 95 to 114 tok/s. Reading 17.9 GB of weights at 300 GB/s bounds plain decode at ~16.8 tok/s, and speculation supplies the rest.

dataroom and searchbox

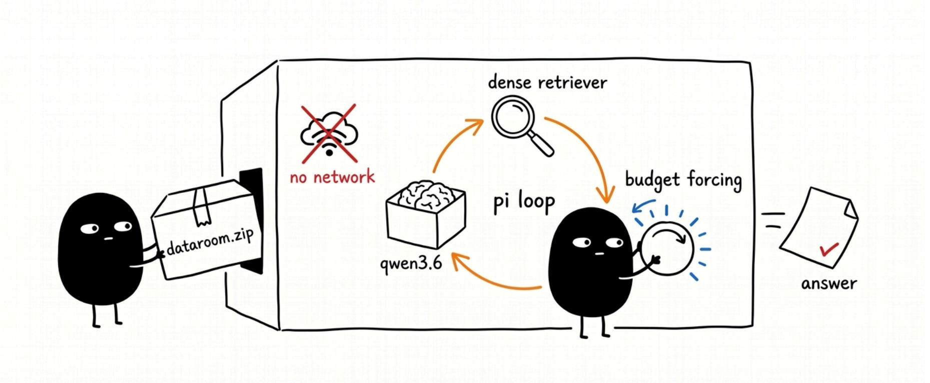
dataroom



A local model runs **search, read and write** in a loop until the knowledge is dumped into one fully cited `.zip` on disk. It stops at a measured coverage floor.

dataroom.jina.ai

searchbox



An **airgapped** testbed for search as test-time compute. One local model is locked in with a single `.zip` dataroom and no web access, so every answer is composed from local tools over what is in the box.

hanxiao.io/searchbox

The private corpus

GROUP	FILES	DETAIL
papers	20	six years of published work, 2023-07 to 2026-07
model-cards	29	one spec sheet per released model
blog	161	tech 104 / press 28 / insights 14 / events 8 / kb 7
api	3	OpenAPI spec, usage guide, model selection guide
faq	2	163 question-answer pairs over 14 sources
site	2	27 page namespaces, 2,297 English strings
legal	1	terms, privacy, security, DPA

Snapshot 2026-08-13, 218 files.

7,777

chunks indexed over 210 documents, median 91 units, longest 195 units at 2.1 KB

768

dimensions, jina-embeddings-v5-text-nano, L2-normalized

Six years of my own papers sit alongside the website material. Each released model carries both a paper and a release post, framed differently, and that overlap is what makes multi-hop questions possible over this corpus.

Verifier construction

Corpus
207-file snapshot of the same material

→

Knowledge graph
a local model extracts grounded triples

→

Path mining
simple entity paths of length 2 and 3

→

Multi-hop question
bridge entities hidden, endpoint is the answer

A path of length k is a chain of k corpus-grounded facts, so path length sets the difficulty. The question names the starting entity and hides the bridges, which forces a solver to discover them. A question enters the set when a closed-book probe fails on it and at least one retrieval-equipped solver reaches the answer, so the set measures search over the corpus. The graphs were built over an earlier 207-file snapshot of the same material, and the 218-file snapshot is what the agentic harness indexes today.

10	**18,615**	**40**	**2 and 3**
graphs built over that corpus by the same local model	edges audited, every edge of every graph	questions sampled per graph	hop counts the question set covers

Evaluation design

	CLOSED-BOOK	SINGLE-SHOT BM25	AGENTIC BM25, 4 ROUNDS
Qwen3.8-27B 104,192 ctx	control that removes items answerable from memory	top-5 passages, one shot	the condition under test
Qwen3.6-35B-A3B 56,320 ctx	control, same corpus	top-5 passages, one shot	the incumbent

HELD FIXED

The corpus, the question set, the retrieval index, the grading procedure and the four-round cap. The agent runs in the airgapped loop with one retrieval tool over `embeddings.npz` and `chunks.jsonl`, plus stock file access. Only the served model changes across rows.

SCORING

The answer held at the end of every turn is captured and graded by the same local judge, so accuracy is reported against turn index and against cumulative fresh-prefill input tokens. Turns are what a person reads and fresh-prefill tokens are what the GPU pays.

A private corpus, a private verifier, and one low-budget GPU decide which model I serve.

Han Xiao
VP of AI, Elastic

@hxiao · in/hxiao87

THE HARNESS